

Prior Design Attempts and Analysis:

Prior Design Attempt #1 - Stall Stopper



Details

This portable lock, known as the Stall Stopper, was designed in order to provide safety and ease of mind when using a restroom where the locks are broken. The Stall stopper costs \$4.75 on its official website. The bright green color paired with the “Occupied” printed on the front stands out to the people outside the stall. It also allows the user not to interact with stall locks.

Analysis

One area of improvement to this design would be how it becomes unusable on stalls where the door does not have a gap on top of the door (ex. Porta-Potty). Another area of concern is the plastic the Stall Stopper is made out of. It is not provided what type of polymer the device is constructed of, so the ability to withstand force is in question. According to Business Insider¹, the lock “slides on the top or bottom of a public restroom door” rather than tightening onto the door in some way, making it easier for the device to get dislodged if disturbed.¹

Prior Design Attempt #2 - <First Watch Security Chrome Travel Lock #2>



image from the same citation in the sources section

This lock can be purchased from the online Home Depot organization as an additional safety measure when traveling. At a price point of \$19.39, this device serves to provide peace of mind for belongings as well as personal well-being by creating a way to lock drawers and doors.

The image illustrates many design choices included in this lock design. For one, the lock operates from the inside using an external key, which the operator must carry with them. This lock is also meant to be operated from the inside of whatever area is being locked, allowing the user to operate the device without a second party involved. This lock is also able to be used on most types of doors, which is an important factor for our criteria of diversified use.

Analysis:

While this design meets the specific needs of some customers, some shortcomings reduce its effectiveness for the problem at hand. One such shortcoming is the necessity of a separate and potentially faulty key to lock and unlock the mechanism. A design with multiple pieces makes it more difficult for the customer, as the key adds an extra element to the mechanism that could become more complicated to operate. Evidence of this can be found in the reviews², where customers complained that “[this] product did not work at all. The key stuck into the lock and then it fell apart...” and “I opened two of the three that I bought and the keys don't work.”

Another potential concern is the cost of the First Watch lock in relation to its quality. Customer reports² stress the instrument's "poor metal quality" This may discourage customers who are looking for an effective and inexpensive solution from buying a \$20 door lock, specifically a lock with negative quality comments where the recipient of the lock is a child and the parent we are working with. A final disadvantage of this product is its design and composition. This lock is made entirely of metal², and has serrated edges that appear somewhat knife-like, however reviews show that the lock "worked for two times then the teeth were rounded and it could slide right off." The combination of lackluster material and shape with minimal structural integrity may pose a problem for our traveling stakeholders.

Prior Design Attempt #3 - RESTROOM STALL BLOCK-AIDE

Image



Details

The portable bathroom lock uses an H shape to slide in between the opening in a bathroom stall door. The lock rests on the pre-installed lock, and it effectively blocks the

stall door from being opened. This lock costs \$29.50 on Equiparts. This design is constructed out of “sturdy ABS resin”³ and appears to be very robust. The “SORRY, OUT OF SERVICE” written on the front of the design humorously acts as a deterrent from bystanders trying to enter the stall.

Analysis

The lock, while being effective on the average bathroom stall, may not work on stall doors that have too small a gap between the wall and the door. The product description notes that the Block-Aide was created “specifically for out of order partition stalls,”³ and that in order to operate, the instrument would have to be able to slide down in a space “between the pilaster and door.”³ With these restrictions in mind, it is important to note that the device may fit into some pockets, but not all. With our stakeholders potentially using restrooms without typical partition stalls, this lock would not be able to be used.

There is also the issue of security that has been brought up by customers on the Equiparts website (sellers of repair parts such as the Block-Aide) asking if there was any way to actually lock or affix the device to the door to prevent the apparatus from “easily sliding³ up.” The company replied and confirmed that the device “...doesn’t stop people from removing [it],”³ calling into question the reliability of the lock.

Prior Design Attempt #4 -BTideas 2 Pack Adjustable Portable Door Lock>

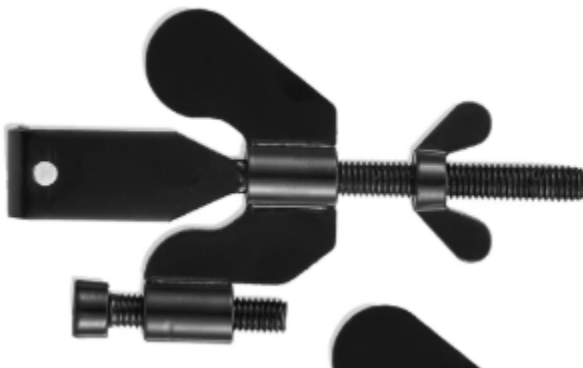


Image credited to cited source #4

This lock was created by the company BTideas and retails through Walmart for \$12.59. This design targets efficiency and travel safety, with some of the main selling points on the website⁴ being sturdiness and portability, specifically mentioning how the lock is small enough to be easily transported in backpacks or luggage, as well as being made of Manganese steel. This portable lock features a locking mechanism that relies on tension from screws and a hook that lodges into the door's internal crevice.

Analysis:

An area to improve upon this past design would be its realm of use. The website⁴ shares that the lock cannot operate on outward swinging doors, sliding doors, or a seamless door. Since our design relies on the criteria of operation on all types of bathroom doors our stakeholders may encounter, the door type restrictions would be a significant issue in this project.

Prior Design Attempt #5 - <Dongyi Portable Door Lock >



Image credited to cited source #5

Details

This portable bathroom lock design by Dongyi acts as a doorstop by utilizing frictional forces to keep doors from getting pushed open or closed, depending on customer preference. The composition of the stopper is reinforced aluminum alloy⁵ with a rubber stopper foot (proclaimed “anti-slip rubber” on the website) that serves as prevention for opening the door.

Analysis

This bathroom lock design, while being effective on doors that reach the floor, would not be usable on a door that does not reach the floor (like most bathroom stall doors). It only works on doors with a 0.2”-2” gap between the floor and the door. It can also only be used on the side that the door opens towards.

Additionally, according to the product warning section on the website, “This product can expose you to chemicals including diethylhexyl phthalate or lead, which are known to the State of California to cause cancer and birth defects or other reproductive harm.”⁵ Since this product is to come in contact with the parent and potentially the child, as well as traveling with them in their backpacks or luggage, using this item put them at risk of harm.

One part of Dongyi’s design that stuck out was the vibrant red color of the door stopper, which allows for high visibility, one of the details in the product description that the stopper was promoted for. This is an element that we should incorporate into our design, as the small child’s comfort relies on their ability to see the device being used to lock their bathroom stall. A brightly colored lock is likely to put the child at ease while simultaneously making the device easy for their parents to locate and use.

Citations

1. (justification) Business Insider, 01-06-2021, “Million Dollar Idea: A Clip That Prevents Restroom Doors From Opening”, Business Insider, www.businessinsider.com/million-dollar-idea-a-clip-that-prevents-restroom-doors-from-opening-2011-1. Accessed 13 Jan. 2026.
2. () Home Depot, No Publication Date, "First Watch Security Chrome Travel Lock 1846",

<https://www.homedepot.com/p/First-Watch-Security-Chrome-Travel-Lock-1846/202799703> Accessed 13 Jan. 2026.

3. () Equiparts, No Publication Date, "RESTROOM STALL BLOCK-AIDE", Equiparts, <https://www.equiparts.com/product/restroom-stall-block-aide/> Accessed 13 Jan. 2026.
4. "BTideas 2 Pack Adjustable Portable Door Lock,Heavy-Duty Hotel Safety Door Lock for Traveling & Home." Walmart.com, 2026, www.walmart.com/ip/BTideas-2-Pack-Adjustable-Portable-Door-Lock-Heavy-Duty-Hotel-Safety-Door-Lock-for-Traveling-Home/18092959304?classType=REGULAR&athbdg=L1700&adsRedirect=true. Accessed 14 Jan. 2026.
5. "Dongyi Portable Door Lock No Tools Install - 230 Lbs Resistant Intruder Defense for Apartment/Travel, Adjustable Fit 0.2-2" Doors,Red (Not Alarm Device)." Walmart.com, 2026, www.walmart.com/ip/Dongyi-Portable-Door-Lock-Tools-Install-230-lbs-Resistant-Intruder-Defense-Apartment-Travel-Adjustable-Fit-0-2-2-Doors-Red-Not-Alarm-Device/18205759840?classType=VARIANT&athbdg=L1400. Accessed 14 Jan. 2026.